CHM111 Exam on chapter(s): 1, 2 100% Excellent!

Name:_ ____

These questions assess your understanding of the material from Chapters 1, 2 of Openstax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

   400 mol ÷ 151. L

   ANSWER: 3 mol / L

2. Provide the name of the element that the element symbol Al represents.

   ANSWER: Aluminum

3. Determine whether magnesium and iodine can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form. Explain why you chose your answer.

   ANSWER: MgI_2

   magnesium typically has a 2+ charge and iodine typically has a 1- charge

4. A geophysicist measures the pressure in a borehole. The pressure is 6.695×10^7 Pa. What is the pressure in megapascals? Report your answer in standard notation, as a decimal.

   ANSWER: 66.95 MPa

   $$6.695 \times 10^7 \text{ Pa} \times \frac{1 \text{ MPa}}{10^6 \text{ Pa}}$$

5. Classify the following as an element, a compound, a homogeneous mixture, or a heterogeneous mixture: seawater filtered until it appears perfectly transparent. You must explain your answer with definitions.

   ANSWER: homogeneous mixture

   because it appears uniform throughout but it is not a compound because it is not pure H_2O

+5

6. Sophia measures the mass of a steel ball bearing using a balance. The accepted mass of the object is 26.7 g. She records a measurement of 27.31 g. What is the percent error of her measurement? Report your answer to two significant figures.

ANSWER: 2.3 %

$$\frac{27.31g - 26.7g}{26.7g} = 0.02285 \times 100\%$$

✓

7. Rachel needs to find the volume of a 63.58 g sample of isopropyl alcohol. The density of the sample is 3.20 $\frac{g}{mL}$. What is the volume of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: 19.9 mL

$$D = \frac{m}{v} \qquad 3.20\ g/mL = \frac{63.58g}{v}$$

$$v = \frac{63.58}{3.20}$$

✓

8. Classify the density of lead as a physical property or a chemical property. Explain your answer.

ANSWER: physical property

because you can measure the density without changing the chemical composition

✓

9. Brian is a thermodynamics researcher who must express a temperature in SI units. He measured the temperature using a Fahrenheit thermometer. The temperature is 158 °F. Convert this temperature to kelvin. Round to the nearest integer.

ANSWER: 343 K

$K = °C + 273.15$

$°C = 5/9\ (°F - 32)$

$°C = 5/9\ (158 - 32)$

$°C = 70.0$

$K = 70.0 + 273.15$

$K = 343.15$

✓

10. Anna must note the concentration of a solution of NH_4NO_3 in her lab notebook. She knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Anna weighs an empty graduated cylinder and finds its mass to be 5.00 g. She then puts some solid NH_4NO_3 into the graduated cylinder and weighs it. With the NH_4NO_3 added, the cylinder weighs 26.6 g. She adds water to the graduated cylinder and dissolves the NH_4NO_3 completely. Then she reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 28.3 mL. What concentration should Anna write down in her lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: 0.763 g/mL

$$\frac{\text{disolved}}{\text{total V}}$$

$26.6g - 5.00g = 21.6g\ NH_4NO_3$ disolved

$$\frac{21.6g\ NH_4NO_3}{28.3\ mL}$$

✓

11. Write the chemical formula for chromic acid.

ANSWER: H_2CrO_4

✓

+6

12. What is the name and symbol of the metric prefix that represents a factor of 10^{-3}?

ANSWER: milli (m)

13. Identify the state of matter in which carbon dioxide exists at 25 °C and 1 atm.

ANSWER: gas

$°F = 9/5(25) + 32$

$= 77 °F$

14. Provide the chemical formula for the molecular compound sulfur dioxide.

ANSWER: SO_2

15. Identify whether the following describes a random error or a systematic error, and explain your answer: A student consistently reads a meniscus from above instead of at eye level, always recording a value that is too low.

ANSWER: systematic error

because the reading is consistently lower and not randomly high or low

16. As part of her duties as an analytical chemist, Hannah measures the amounts of elements E_1 and E_2 found in four samples collected from an unknown substance X. She then records these masses in a table in her lab notebook, reproduced below. It is known that X contains no elements other than E_1 and E_2. Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E_1 that Hannah would find in a new 10.0-g sample of X. Round your answer to two significant figures. Explain why you chose your answer.

sample	mass of E_1		mass of E_2	
1	12.9 g	÷	7.4 g	= 1.7
2	11.7 g	÷	6.7 g	= 1.7
3	6.8 g	÷	3.9 g	= 1.7
4	7.5 g	÷	4.3 g	= 1.7

$1.7 + 1 = 2.7$

$\frac{1.7}{2.7} = 0.63$

$0.63 \times 10 g$

$= 6.3$

ANSWER: pure substance, 6.3 g E_1

you know it's a pure substance because the ratio between the elements stays the same throughout samples

+5

17. Provide the number of protons and the number of neutrons there are in an atom of ^{31}P.

ANSWER: protons = 15 (atomic number)
neutrons = 16 (mass # - atomic #)

18. Classify luster as an intensive or extensive property. You must explain your answer using the definition of the properties.

ANSWER: Intensive property
because the amount of a substance you have does not determine luster

19. Sarah measures the speed of a moving object and records 88.6 $\frac{km}{h}$. Convert this speed to $\frac{m}{s}$. Ensure your answer has the correct number of significant figures.

$$\frac{88.6\ km}{1\ h} \times \frac{1{,}000\ m}{1\ km} \times \frac{1\ h}{60\ min} \times \frac{1\ min}{60\ s}$$

ANSWER: 24.6 m/s

20. Answer the following questions: is F an atom, a cation, or an anion? How many protons does it have? How many electrons does it have?

ANSWER: atom, p = 9, e = 9
because no charge is stated meaning it is neutral

21. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

2.1385 cm – 0.56531 cm

ANSWER: 1.5732 cm

+5

22. Provide the name and empirical formula of the ionic compound that is formed by the NH_4^+ cation and the PO_4^{3-} anion.

ANSWER: $(NH_4)_3PO_4$, ammonium phosphate

23. State the SI base unit used to measure amount of substance and give its symbol.

ANSWER: mole (mol)

24. Classify S_2Si as either an ionic or a molecular compound. Explain why you chose your answer.

ANSWER: molecular compound

because it consist of 2 nonmetals; an ionic compound must have a metal

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1. $3\ \frac{\text{mol}}{\text{L}}$
2. Aluminum
3. Yes, magnesium and iodine form an ionic compound with the empirical formula MgI_2.
4. 66.95 MPa
5. It is a homogeneous mixture. ∵ it is a combination of two or more substances with a uniform composition and appearance.
6. 2.3% ∵ % Error = $\left|\frac{E-A}{A}\right| \times 100\%$
7. 19.9 mL
8. Physical property. The density of lead is a physical property; it can be measured without changing the chemical identity of lead.
9. 343 K
10. $0.763\ \frac{\text{g}}{\text{mL}}$
11. H_2CrO_4
12. Milli (symbol: m)
13. Gas
14. SO_2
15. Systematic error ∵ the incorrect technique consistently shifts every measurement in the same direction.
16. X is a pure substance; Mass of E_1: 6.4 g
17. 15 protons and 16 neutrons.
18. Intensive property. Luster is an intensive property ∵ the shininess of a substance does not depend on the amount present.
19. $24.6\ \frac{\text{m}}{\text{s}}$
20. F is a neutral atom. It has 9 protons and 9 electrons.
21. 1.5732 cm
22. Ammonium phosphate: $(NH_4)_3PO_4$
23. The SI base unit for amount of substance is the mole (symbol: mol).
24. S_2Si is a molecular compound.